KARTIKEYA VEMULA

Data Scientist | Experimentation & ML Systems | Time Series Forecasting | Predictive Analytics
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PROFILE

Data Scientist specializing in **experimentation systems, predictive modeling, and time series forecasting** for data-driven product decisions. Experienced building **statistically rigorous A/B frameworks, anomaly detection pipelines, and production-style ML systems** from validation to stakeholder rollout.

CORE COMPETENCIES

Predictive Modeling & ML Systems | Statistical Experimentation & A/B Testing | Time Series Forecasting
Causal Inference | Selection Bias Mitigation | Feature Engineering | Model Evaluation | Stakeholder Communication

TECHNICAL SKILLS

ML & Modeling: Python (scikit-learn, XGBoost, Random Forest, Ensemble Methods), SHAP Explainability, Cross-Validation **Experimentation:** A/B Testing, Z-test and T-test Inference, Power Analysis, SRM Detection, Causal Inference, FDR Control **Time Series & Forecasting:** ARIMA (Box-Jenkins, ADF, ACF/PACF), Meta Prophet (changepoint tuning), Walk-Forward Backtesting (RMSE, MAE) **Advanced Modeling:** Extreme Value Theory (GPD), Ensemble Methods, Bayesian Inference, Anomaly Detection
Data Engineering: SQL, ETL Pipelines, BigQuery (distributed processing), Apache Spark, REST APIs, Star Schema Design **Visualization & Tools:** Tableau, Power BI, Plotly, Matplotlib, Git, GitHub, VS Code, YAML Orchestration

PROJECTS

AB-Engine: Production-Style Experimentation Platform with Metric Strategy Layer | Python | NumPy | SciPy | Pandas | YAML | 2025 • Defined **primary and secondary success metrics** across **10,000 simulated users**, aligning short-term conversion lift with long-term retention signals to shape experiment prioritization and scope.

- Implemented **deterministic SHA-256 bucketing**, power analysis, **Z-tests**, **FDR controls**, and **SRM detection** within a unified experimentation pipeline. • In a checkout flow optimization experiment, detected a **+3.4% conversion lift (p = 0.038)**. Guardrail logic simultaneously flagged a **1.8% retention decline in the Mobile/iOS segment**, triggering a localized rollback despite overall positive primary lift.
- Implemented **selection bias mitigation** and guardrail monitoring to preserve causal validity across multi-segment user populations. • Auto-generated **plain-language rollout recommendations** via YAML-driven config, enabling product stakeholders to act on results without SQL or statistics expertise.

Predictive KPI Monitoring & Root Cause Diagnostic Engine | Python | SQL | XGBoost | BigQuery | Plotly | 2025

- Built an **XGBoost-driven anomaly detection pipeline** on **~1.2M KPI observations** across 3 product dimensions (Device, Region, Browser), with a **100/100 data quality gate** running null-rate and volume checks before any model inference.
- XGBoost predicted KPI residuals using **12 engineered features** (day-of-week, seasonal lags, rolling averages); tuned via **5-fold time-series cross-validation**. Rolling Z-score baseline achieved **88% recall but only 62% precision** due to seasonal over-sensitivity; XGBoost improved precision to **84% and recall to 91%**.
- SHAP analysis identified the **top 3 features explaining 71% of anomaly variance**. Automated segment-level root cause attribution cut analyst investigation time by **35%**, with ranked JSON diagnostics surfaced directly to stakeholder dashboards.

Extreme-Value AQI Prediction & Early Warning System | XGBoost | Random Forest | GPD | SHAP | 17 Years EPA Data | 2025 • Developed a **rare-event forecasting system** for hazardous AQI events using **17 years of EPA data (~2.1M observations)**, applying **Extreme Value Theory (Generalized Pareto Distribution)** to model tail-risk pollution spikes that standard ML assumptions underfit.

- Engineered **17 temporal, seasonal, and pollutant-level features**; SHAP identified **PM2.5 and NO2 as primary drivers** of extreme events. Ensemble (XGBoost + Random Forest) achieved **RMSE 9.8 and R^2 0.84** vs. linear regression baseline R^2 0.61.
- Translated outputs into **interpretable risk indicators** and **72-hour early warning signals**, enabling pre-emptive public health advisories for policy-level stakeholders.

Multi-Asset ARIMA Forecasting & Anomaly Detection Pipeline | Python | ARIMA(2,1,2) | Meta Prophet | statsmodels | ADF | 2025 • Built a **per-asset ARIMA(2,1,2) pipeline** across **3 assets (NVDA, AMD, MSFT)** with **~1,825 daily observations per asset**, applying **ADF stationarity testing** and **ACF/PACF-driven lag selection** via Box-Jenkins methodology. All assets required d=1 differencing.

- Compared **ARIMA vs Meta Prophet** across asset classes: Prophet achieved **12% lower RMSE on high-seasonality equities (NVDA, MSFT)** versus naive ARIMA(1,0,0) baseline; ARIMA outperformed on mean-reverting series. Generated **30-day forecasts with 95% confidence intervals**. • Validated through **walk-forward backtesting** with AIC/BIC model selection and residual white-noise checks, improving detection precision by **25% versus fixed-threshold baseline**.

Engineering Analytics Data Warehouse & Insight Pipeline | GitHub REST API | SQL | ETL | BigQuery | Star Schema | 2024 • Designed a **production-grade ETL pipeline** ingesting data across **50+ repositories** into a **BigQuery-compatible star schema** with clean fact and dimension table separation, engineered for scalable distributed workloads.

- Implemented **idempotent ingestion logic** with defensive handling of pagination failures, rate-limit breaches, and malformed payloads, eliminating silent data corruption across **500+ repeated pipeline runs**.
- Delivered **SQL-based contributor analytics** covering commit velocity, repository health, and team productivity patterns, enabling data-driven decisions surfaced directly to engineering leadership.

EDUCATION

M.S. Data Science | University of Maryland Baltimore County, USA | 2024 to 2026
GPA: 3.78 / 4.0 | Focus Areas: Machine Learning, Statistical Modeling, Experimental Design, Time Series Analysis

B.Tech Computer Science | GITAM University, Visakhapatnam, AP, India | 2020 to 2024
GPA: 8.89 / 10 | First-Class with Distinction

CERTIFICATIONS

Applied ML in Python (University of Michigan) | **Neural Networks & Deep Learning** (DeepLearning.AI) | **Data Analysis with Python** (IBM)